TRANSFINITTE · HARDWARE TRACK · PROTOTYPE BUILD GUIDE · V2 (24 OCT 2026)

HridayMitra: requirements, parts list and complete build guide

Monitoring only, not a diagnosis tool. Never test on a person with a pacemaker or defibrillator. Circuit values are starting values: check them in LTspice before you build.

What changed in v2

- **Lung fluid is now measured physically.** The 4 hand strips are wired as electrodes. Two 74HC4052 switches pick 4 current paths (legs, arms, right side, left side), and the ESP32 computes a **trunk (chest) impedance**. This replaces the simulation-only chest belt. New rows and wiring are marked in yellow.
- **Prices revised from Indian retail listings** (Probots, Robocraze, Hubtronics, Robu, Thingbits, Zbotic, IndiaMART), searched on 24 Oct 2026. ✓ = found online; the rest are estimates.
- **The ESP32 is already bought.** It costs nothing more to buy, but its market price (₹499) is counted in the product parts list.
- **Instrumentation amplifier fixed.** The INA128 and AD620 need at least about 4.5 V, so they won't run on our single 3.3 V supply. Use the INA126 (SOIC, on an adapter) or an INA333 breakout. The fallback is a 3-op-amp version built from 2 × MCP6002.
- Battery, voice and button prices went up; the totals are recomputed in section 3.

1. What the PS requires (our checklist)

Rule	How the prototype meets it
Under 2 min a day, no all-day patch	A seated check of about 75–90 s, plus 20 s in the evening. The amulet is worn like a pendant, not stuck on
Local language, little reading	Voice prompts, 3 lights, big buttons
Power cuts, weak internet	Battery, offline logic on the ESP32, SMS over 2G (not on Jio)
Parts under ₹5,000	Product parts list: **₹4,436 at the lowest prices found, up to ₹7,095 at planning prices** (section 3). The rule is met only if we buy near the low prices. Get real quotes

Warning sign	Measured by
Weight gain	4 load cells under the plate
Faster resting heart rate	MAX30102 finger clip (ECG optional)
Faster breathing	Amulet motion sensor at night (experimental), plus symptom questions
Less activity	Amulet steps, stand-up speed from the load cells, missed check-ins
Fluid in legs	Foot-to-foot impedance at 5 kHz (50 kHz in simulation), trended against the patient's own baseline
Fluid in lungs	**Trunk (chest) impedance from 4 hand + 4 foot electrodes**: $Z_{trunk} = (Z_{right\ side} + Z_{left\ side} - Z_{arms} - Z_{legs}) \div 2$, trended against the patient's baseline. Supported by SpO_2, symptoms and night sit-ups
Trouble breathing lying flat	Amulet posture and night sit-ups, plus the pillow question

Deliverables: user journey; block diagram; schematic of the impedance circuit with the ESP32; parts list in INR; simulation proof (LTspice, Wokwi, Python); code with a personal baseline and a reason for every alert; a simple status screen; 200+ synthetic patients with sensitivity, specificity, median lead time and false alerts per patient-month on a patient-level split; BIDMC heart-rate test; public GitHub with setup steps; 2-page report; 5-minute pitch; the "not a diagnosis tool" disclaimer.

Judging: sensing creativity 20, ease and long-term use 20, software 20, hardware and simulation 15, science and honest testing 15, cost, India fit and pitch 10.

2. What is physical, and what is simulated

Physical (build)	Simulated or shown on screen only

- Stool on a plywood plate with 4 load cells (weight)
- Finger pulse and SpO_2
- 3 lights and 3 buttons
- **Leg and trunk (lung) impedance at 5 kHz**: resistor phantom test first, then real foot plates and hand strips
- Laptop running the alert engine
- Amulet with motion sensor over Bluetooth, if time
- SMS by SIM800L, if time

- Impedance at 50 kHz and the full circuit (LTspice)
- The 200+ synthetic patients and all metrics
- The break-it demo
- The family and ASHA message sequence (phone mock-up)

Block flow: electrodes and sensors → ESP32 (readings, quality checks, lights, buttons) → USB serial JSON → laptop alert engine → state back to the ESP32 lights, SMS to a phone.

3. Complete parts list (revised prices)

Tiers: **Must** = needed for the tier-A demo. **Should** = brownie points if time allows. **Nice** = polish. **Own** = you have it. Prices are in ₹: **Low** = cheapest listing found, **Plan** = a safe retail price. ✓ = found online on 24 Oct 2026 (prices and stock change); unmarked = estimate. Yellow rows are new or changed in v2.

#	Part and values	Qty	Used for	Tier	Low ₹	Plan ₹	Source
A. Parts you already own (₹0 to buy; their market price is counted in the product parts list)							
1	ESP32 WROOM-32 DevKit (classic). **Already bought.** Has the DAC used for the impedance signal	1	Station brain	Own	0	0	Market ₹499 ✓ (Amazon.in, tracked)
2	AD8232 ECG module. Optional rhythm check from the hand electrodes. Not in the product parts list	1	Bonus	Own	0	0	Market ₹443–528 ✓
3	ESP32-S3 Sense. Demo stand-in for the amulet brain; turn the camera off	1	Amulet (demo)	Own	0	0	Product uses ESP32-C3 SuperMini ₹230 ✓
4	Buttons, joystick. Use 3 push buttons: green, red, SOS	3	Answers, SOS	Own	0	0	
B. Sensing and signal chain							
5	50 kg bar load cell + HX711 board, sold as sets. Use all four; each cell has its own HX711 and the software adds the four weights	4	Weight	Must	544	880	Robocraze ₹136/set ✓; Engineer Store ₹220 ✓
6	MAX30102 module, I2C 0x57, 3.3 V	1	Pulse, SpO_2	Must	130	139	Hubtronics ✓; Probots (in stock) ✓
7	**INA126 (SOIC-8) on a SOIC-to-DIP adapter**, single 3.3 V, gain 10 with RG 16 kΩ. Alternative: INA333 breakout. **Do not use INA128 or AD620** (they need about 4.5 V or more). Fallback: 3-op-amp version from 2 × MCP6002 + 1% resistors	1	Reads body voltage	Must	154	265	Robu INA126 ₹139 ✓ (out of stock); adapter ₹15
8	Dual op-amp, DIP-8: MCP6022 (rail-to-rail, 10 MHz) or **MCP6002** (fine at 5 kHz)	1	Vref and excitation buffer	Must	46	80	Hubtronics MCP6002 ₹46 ✓; MCP6022 not found in India
9	74HC4053 analog switch, DIP-16 (CD4053 works too). Selects body voltage or current-sense voltage	1	Measurement select	Must	22	30	Probots CD4053 ✓; 74HC family ~₹30 ✓
9b	**74HC4052 dual 4-channel analog switch, DIP-16** (CD4052 works). U4 picks the current pair, U5 picks the sense pair: legs, arms, right side, left side	2	Lung-fluid paths	Must	44	60	Estimate (same family as CD4051 ₹22 ✓)
10	DIP sockets: 2 × 8-pin, 3 × 16-pin	5	Easy part swaps	Must	15	35	
11	Resistors, 1% metal film, ¼ W: 10k ×5 (incl. 2 boot pulls), 100k ×1, 4.7k ×1, 220R ×4 (R5 + 3 LEDs), 16k ×1 (or 15k + 1k), 1M ×2, 1k ×6, 1.5k ×1, 2.2k ×1. Phantom: 330R ×4, 47R ×1, 33R ×1, 680R, 1k	~32	Impedance circuit, LEDs, phantom	Must	60	250	
12	Capacitors: 1 µF X7R or film (16 V+) ×4, 100 nF ×8, 10 µF ×3, 10 nF ×2, 100 pF ×2 spare, 1000 µF 10 V electrolytic ×1	~20	Coupling, decoupling, filters	Must	60	150	
13	Diodes: 1N5819 Schottky ×2 (reverse-polarity protection), 1N4148 ×4 spare	6	Protection	Must	10	25	
14	LEDs 5 mm: green, amber, red (with the 220R resistors above)	3	Status lights	Must	10	25	
C. Frame, electrodes, wiring							

15	Stool: any sturdy flat-top stool, seat about 40–45 cm. Foldable all-in-one is the product design	1	Body	Must	250	550	Local
16	Plywood plate, about 40 × 40 cm, 12–18 mm thick	1	Weighing platform	Must	80	150	Local
17	M5 bolts, nuts, washers: M5 × 30 mm ×8, nuts ×8, washers ×16, plus 5 mm spacers or 3 mm plywood strips	1 set	Mount load cells	Must	40	80	Local
18	Rubber feet, under the free end of each load cell	4	Floor contact	Must	20	40	Local
19	PVC pipe 25 mm, 1 m, with flange and elbow	1 set	Handle post	Must	100	200	Local
20	Stainless-steel (304) strips: 4 foot plates about 8 × 5 cm and **4 hand strips about 10 × 2 cm, now all wired**. Round all edges	8	Electrodes (legs + lungs)	Must	200	320	SS304 ₹190–250/kg ✓ (IndiaMART) + cutting
21	Shielded 4-core cable: about 3 m for the feet **+ 2 m up the post for the hands**	2	Electrode wiring	Must	80	140	
22	Hookup wire 22 AWG: red, black, plus 2 colours, about 3 m each	1 set	Wiring	Must	40	80	
23	Project box about 150 × 100 × 50 mm, fixed under the seat	1	Electronics enclosure	Must	60	100	
24	Breadboards, jumper wires: 2 × 830-point, M-M, M-F and F-F sets	1 set	Prototyping (demo only)	Must	120	200	
25	Screws, M3 standoffs, cable ties, heat-shrink, tape, velcro, hot-glue sticks	1 set	Assembly	Must	60	110	
26	Solder, flux (if you don't already have them)	1	Soldering (demo only)	Must	0	40	
27	5 V 2 A phone charger and cable (skip if you own one)	1	Power	Must	0	150	Counted at ₹150 in the product list
28	Anti-slip mat on the foot area	1	Safety	Nice	30	50	
D. Messaging, voice, battery							
29	SIM800L GSM module + antenna. 2G; needs 3.4–4.4 V and 2 A bursts. Use an Airtel, Vi or BSNL SIM (not Jio)	1	SMS and calls	Should	150	448	Verified range ✓ (Zbotic, IndiaMART)
30	18650 cell (protected), holder, TP4056, MT3608 boost, slide switch	1 set	Battery, charge, 5 V rail	Should	225	381	Cell ₹96–199 ✓, holder ₹69 ✓, TP4056 ₹49 ✓, MT3608 ₹35–49 ✓
31	DFPlayer Mini + microSD (2–4 GB) + **4 Ω 3 W speaker** (louder for elderly users). For a first demo, play voice from the laptop	1 set	Local-language voice	Should	391	518	DFPlayer ₹72–119 ✓, speaker ₹199–249 ✓ (Probots), SD estimate
32	Perfboard 7 × 9 cm + header pins, to move the circuit off the breadboard	2	Permanent circuit	Should	80	150	
33	Large 30 mm arcade buttons, elder-friendly	3	Answers, SOS	Nice	237	280	Thingbits ₹79–93 each ✓
E. Amulet (demo)							
34	MPU6050 breakout (or LIS2DW12). The LIS3DH chip is listed obsolete; check stock	1	Steps, posture, falls	Should	120	199	IndiaMART ₹120 ✓; Probots ₹199 ✓
35	LiPo 3.7 V 400 mAh with protection; check that the connector fits your board	1	Power	Should	120	180	200 mAh: Robocraze ₹124 ✓
36	TP4056 USB-C charge module (skip if the board already charges LiPo)	1	Charging	Should	49	49	Probots ✓
37	Vibration motor + BC547 + 1k + 1N4148	1 set	Haptic prompt	Nice	30	50	
38	Pendant case + cotton thread. Breakaway thread, no magnets (the night-clip version is still being decided)	1	Shell	Nice	60	120	
39	Pogo pins + contact board. Charging dock, no magnets	1 set	Dock	Nice	40	70	
F. Test rigs (not product parts, not counted)							
40	SG90 servo, as a breathing dummy for the amulet test	1	Test	Test	100	150	

41	1 L water bottles: 5 bottles = 5 kg calibration weight	5	Calibrate weight	Test	0	0	
42	Protractor, multimeter: angle checks and voltage checks	1 set	Test	Test	10	400	

Totals

What	Low ₹	Plan ₹	Note
To buy: Must	2,145	4,099	
To buy: Should	1,135	1,925	
To buy: Nice	397	570	
To buy: Must + Should	3,280	6,024	
To buy: everything	3,677	6,594	Test rigs not counted
Product parts list (for the report and the ₹5,000 rule)	**4,436**	**7,095**	**All product parts at market price, including the owned ESP32 (₹499) and ESP32-C3 for the amulet (₹230–242). Excludes breadboards, solder, AD8232, the S3 Sense and test rigs**

Reading the totals honestly. The ₹5,000 rule is met only at prices close to the **Low** column (₹4,436, ~₹560 headroom). The **Plan** column is a padded upper bound. The big swings are:

- load cells (₹544 → 880)
- stool (₹250 → 550)
- SIM800L (₹150 → 448)
- resistors (₹60 → 250)
- battery set (₹225 → 381)

To stay under ₹5,000: buy each part at the Low source above, buy resistors and capacitors as one small kit, play voice from the laptop for the demo (−₹391), and use a stool you already own (−₹250). The planned cost-trim step comes last. Every ₹ in the report needs a cart screenshot or a link.

4. Pin map and wiring (classic ESP32 DevKit)

This map is a proposal; check it against your board's pinout. Boot pins:

- **GPIO12** must be low at power-up. It gets a 10 kΩ pull-down.
- **GPIO0** must be high at power-up. It gets a 10 kΩ pull-up.
- **GPIO15** (SOS button) idles high.

The 74HC4052 select inputs don't drive these pins, so the board still boots normally. Analog inputs use ADC1 pins so they work with Bluetooth on.

Function	ESP32 pin	Wiring notes
HX711 data ×4	GPIO32, 33, 26, 27	One HX711 per load cell, powered from 3.3 V
HX711 clock (shared)	GPIO14	Same clock to all four boards
Excitation (DAC)	GPIO25	5 kHz cosine, amplitude scale ½
Impedance read (ADC)	GPIO35	From the INA126 output filter
Measurement select	GPIO18	74HC4053: low = body, high = current sense
Path select S0	GPIO12	To A pins of U4 and U5 (74HC4052). 10 kΩ pull-down to GND (boot strap)
Path select S1	GPIO0	To B pins of U4 and U5. 10 kΩ pull-up to 3.3 V (boot strap). If flashing or booting fails, play voice from the laptop and move S1 to GPIO19
I2C SDA, SCL	GPIO21, 22	MAX30102 (0x57)
ECG OUT, LO+, LO−	GPIO34, 39, 36	AD8232, optional
LEDs green, amber, red	GPIO13, 23, 2	Each through 220 Ω to GND
Buttons green, red, SOS	GPIO4, 5, 15	To GND, using internal pull-ups

SIM800L TX/RX (UART2)	GPIO17, 16	ESP32 TX → 1 kΩ → SIM800L RX, with 2.2 kΩ to GND (about 2.3 V). Power the SIM800L from the battery with a 1000 µF capacitor, not from the 3.3 V pin
DFPlayer TX	GPIO19	UART1 remapped. ESP32 TX → 1 kΩ → DFPlayer RX; DFPlayer on 5 V; speaker on its SPK pins

5. The impedance front end (the custom circuit): every value

Idea: a 5 kHz sine goes through the body via two outer electrodes, and two inner electrodes measure the voltage. A small resistor (R5) in the current path measures the current. The ESP32 reads both through the same amplifier, so **Z = R5 × (voltage across the body ÷ voltage across R5)** and the gain cancels. The bias point (Vref) is 1.65 V, half of 3.3 V. **New in v2:** two 74HC4052 switches choose which electrodes carry current and which sense, so the same circuit measures four body paths.

Electrode names

Name	Where	Role
LF_FORE, RF_FORE	Forefoot plates, left and right	Current
LF_HEEL, RF_HEEL	Heel plates, left and right	Sense
LH_OUT, RH_OUT	Outer hand strips (towards the fingers), left and right	Current
LH_IN, RH_IN	Inner hand strips (towards the wrist), left and right	Sense

The four paths (S1 S0 on GPIO0, GPIO12)

S1 S0	Path	Current in → out	Sense +, −	Measures
0 0	Z1 legs	LF_FORE → RF_FORE	LF_HEEL, RF_HEEL	left leg + right leg (leg fluid)
0 1	Z2 arms	LH_OUT → RH_OUT	LH_IN, RH_IN	left arm + right arm
1 0	Z3 right side	RH_OUT → RF_FORE	RH_IN, RF_HEEL	right arm + trunk + right leg
1 1	Z4 left side	LH_OUT → LF_FORE	LH_IN, LF_HEEL	left arm + trunk + left leg

Trunk (lung-fluid proxy): Z_{trunk} = (Z3 + Z4 − Z1 − Z2) ÷ 2. Fluid in the chest lowers Z_{trunk}. This is the segment model used by 8-electrode body-composition scales. **It is untested for heart failure. The trunk is only about 20–50 Ω, while each path is a few hundred ohms, so the subtraction magnifies noise: trend it over days, never one reading.**

Circuit blocks

Block	Parts and values	Connections
Vref 1.65 V	R1 10 kΩ 1%, R2 10 kΩ 1%, C1 10 µF, U1A (op-amp) as a buffer	3V3 → R1 → node → R2 → GND; C1 from node to GND; node to U1A + input; U1A output tied to its − input = VREF
Excitation	C2 1 µF, R3 100 kΩ, U1B buffer	GPIO25 → C2 → EXC_IN; R3 from EXC_IN to VREF; EXC_IN to U1B + input; U1B output = EXC_OUT (output tied to its − input)
Current path + U4 (74HC4052, current pair)	C3 1 µF, R4 4.7 kΩ 1%, C4 1 µF, R5 220 Ω 1%	EXC_OUT → C3 → R4 → U4 X-common. U4 X0 = LF_FORE, X1 = LH_OUT, X2 = RH_OUT, X3 = LH_OUT. U4 Y0 = RF_FORE, Y1 = RH_OUT, Y2 = RF_FORE, Y3 = LF_FORE. U4 Y-common → C4 → RET → R5 → VREF. U4 A = GPIO12, B = GPIO0, INH = GND, VEE = GND, VCC = 3.3 V, 100 nF
Voltage sensing + U5 (74HC4052, sense pair)	1 kΩ protection on each of the 4 sense electrodes; R8, R9 1 MΩ bias	LF_HEEL, RF_HEEL, LH_IN, RH_IN each → 1 kΩ → U5. U5 X0 = LF_HEEL, X1 = LH_IN, X2 = RH_IN, X3 = LH_IN; X-common = SA. U5 Y0 = RF_HEEL, Y1 = RH_IN, Y2 = RF_HEEL, Y3 = LF_HEEL; Y-common = SB. R8: SA to VREF; R9: SB to VREF. U5 A/B shared with U4, INH = GND, VEE = GND, VCC 3.3 V, 100 nF
Selector (74HC4053)	Two of its three switches; both select pins on GPIO18; third switch unused (select to GND); enable to GND; VEE to GND; VCC 3.3 V; 100 nF	Switch X: X0 = SA, X1 = RET, common → INA + input. Switch Y: Y0 = SB, Y1 = VREF, common → INA − input. Select low = body, high = across R5
Amplifier (INA126)	RG 16 kΩ (15 kΩ + 1 kΩ), gain about 10, 100 nF	REF pin to VREF; V+ 3.3 V; V− GND; RG between its two RG pins; output → R11 1.5 kΩ → GPIO35, with C5 10 nF from GPIO35 to GND
Decoupling	100 nF at each IC supply pin, 10 µF on the 3.3 V rail	Close to the chips

Check pin numbers on the datasheets (INA126, MCP6022/MCP6002, 74HC4053, 74HC4052) before wiring. The 74HC4052 switch resistance (about 100 Ω at 3.3 V) sits in the current loop; it doesn't affect the result, because the current is measured across R5. All switched signals stay between 0 and 3.3 V because they are centred on VREF.

Expected levels (estimates): current about 50–100 µA peak. Across the legs about 35–70 mV peak, so 0.35–0.7 V after the amplifier. Across R5 about 11–22 mV, so 110–220 mV after the amplifier. If the R5 reading is too small, raise R5. If the body

reading clips, lower the excitation scale.

Timing: 4 paths × 2 select states × about 100 ms, plus 0.2 s settling after each path change, × 3 repeats (median) = about 4–5 s, inside the 60 s sit.

Resistor-phantom test (do this before any person touches it)

1. **Legs only:** put 330 Ω between LF_FORE and RF_FORE, with LF_HEEL on the LF_FORE end and RF_HEEL on the RF_FORE end. Path Z1 should read 330 Ω.
2. Repeat with 680 Ω and 1 kΩ. The computed Z should follow the resistor within about 5%. If it doesn't, check the select lines, the bias resistors and the ADC sampling.
3. Add 1–5 kΩ in series with each outer lead to mimic poor contact. Z should barely move, because the current is measured.
4. **Whole-body phantom (new).** Build a star of five resistors:
   - left arm 330 Ω from the left-hand node to a "chest" node
   - right arm 330 Ω from the right-hand node to the chest node
   - trunk 47 Ω from the chest node to a "pelvis" node
   - left leg 330 Ω from the pelvis to the left-foot node
   - right leg 330 Ω from the pelvis to the right-foot node

   Clip each hand's OUT and IN to its hand node, and each foot's FORE and HEEL to its foot node. Expect Z1 ≈ 660, Z2 ≈ 660, Z3 ≈ Z4 ≈ 707, so Z_{trunk} ≈ 47 Ω. Then change the trunk to 33 Ω ("more chest fluid") and check that the computed trunk drops by about 14 Ω. Report the error.

Digital measurement: sample GPIO35 at 40 kHz or more (ADC continuous mode). Take a Goertzel (a fast single-frequency filter) at 5 kHz over about 100 ms for each select state, and compute the amplitude ratio. If the ADC is too slow, drop the excitation to 2 kHz. Let it settle for about 1 s after switching the DAC on. Use the median of three readings.

LTspice proof (graph for the judges): model each segment (arm, trunk, leg) as outside-fluid resistance in parallel with inside-fluid resistance plus membrane capacitance, and add 1–10 kΩ of contact resistance on each electrode. Sweep **leg fluid ±10% and trunk fluid ±10% separately**, at 5 kHz and 50 kHz. Show that leg fluid moves Z1, trunk fluid moves Z_{trunk}, and contact resistance barely moves either. Report the error.

6. Build guide, in order (with checks)

Step	Do	Done when	Time
1. Bring-up	Power the ESP32 from USB, blink the 3 LEDs, read the 3 buttons on serial. **Fit the GPIO12 pull-down and GPIO0 pull-up and confirm the board still flashes and boots**	Each LED lights, each button prints, and it reboots cleanly	20 min
2. Weight platform	Bolt each load cell between the plywood plate and a rubber foot, with spacers so the free end can flex. Wire each cell to its own HX711 (3.3 V, GND, its data pin, shared clock). Read all four and add them. Calibrate with 1, 2 and 5 L water bottles (1 L ≈ 1 kg), store the scale factor, tare on boot	5 kg reads 5 kg within about 0.1 kg at the center and at each corner	40 min
3. Pulse sensor	MAX30102 on I2C (3.3 V); read heart rate with a library example	Heart rate matches your wrist pulse	15 min
4. Serial protocol	Print one JSON line per check: weight, leg impedance, **trunk impedance**, pulse, SpO_2, button answers, quality. Receive the state (green, amber, red) from the laptop and set the LEDs	The laptop script prints each line, and a test state lights the LEDs	20 min
5. Impedance board	Build the section 5 circuit on a breadboard, one block at a time: Vref, then excitation, then selector and amplifier, **then U4 and U5**. Run phantom tests 1–4	Legs phantom within about 5%; whole-body phantom gives trunk 47 → 33 Ω within ±5 Ω	90 min
6. Frame	Fix the stool and the PVC post to the plate (bolts or flange). Fix the handle bar. **Wrap 2 hand strips per hand (OUT towards the fingers, IN towards the wrist, gap of 3 cm or more) and run the 4-core cable down the post.** Fix foot plates with a 2 cm gap between forefoot and heel plates. Insulate the metal from the frame. Mount the project box under the seat; cable ties for the leads	Nothing wobbles; no metal touches another electrode	40 min
7. Electrodes live	Connect the foot plates and hand strips to the impedance board. Test on your own dry bare feet and dry hands, **only after the phantom tests pass and while on battery power**	Stable Z1–Z4. Z1 changes when you wet your feet or lift a heel. Z2 changes when you loosen your grip	30 min
8. Integration	Run the laptop alert engine on the serial stream. Walk through green, amber and red using the break-it demo scenarios	The LEDs follow the engine	30 min

9. Extras	Only if time is left: SIM800L SMS (AT+CMGS), DFPlayer voice, ECG on the hand electrodes, amulet	An SMS reaches a phone	As time allows

7. Firmware and data

Modules: sensors (weight, impedance on 4 paths, pulse), quality checks (stillness, contact, signal), identity check (weight and impedance fit the patient), serial output, LED and button handler, and optional SMS and voice.

Daily features file (the contract between hardware, data and engine): `patient_id, day, weight_kg, leg_z_ohm, leg_z_evening_ohm, arm_z_ohm, trunk_z_ohm, pulse_bpm, spo2, standup_s, breathless, orthopnea, took_pills, amulet_steps, amulet_situps, quality`. A missing value is an empty cell, never 0.

Quality rules:

- Reject weight unless the load is steady for 5 s.
- Reject an impedance path if its select-high reading is below a floor (no current flowing).
- **Reject the trunk value if any of Z1–Z4 failed, or if a hand lost grip** (Z2 jumps above 3× its baseline).
- Reject pulse with a low quality score.

Show "no reading" instead of guessing.

Tare: zero the empty stool on every power-up. Re-check with a 5 kg known weight weekly.

8. Amulet prototype (if time)

1. Wire the MPU6050 to the S3 Sense: 3.3 V, GND, SDA, SCL. Read acceleration at 25–50 Hz.
2. Compute the tilt angle from gravity: lying flat, propped, upright. Count a "sit-up" as upright for more than 5 minutes between midnight and 5 am (shorten the time for the demo).
3. Count steps and movement minutes; flag worn or not worn from tiny movements.
4. Send a Bluetooth notification with angle, steps, sit-up flag and the SOS press.
5. Test angles with a protractor jig (within 5 degrees). Test breathing only with the servo breathing dummy (target: within 2 breaths a minute in at least 80% of windows), and report it as experimental.

Pendant vs night-clip mounting is still an open team decision. A free-hanging pendant swings and slides, which weakens the posture and breathing readings.

9. Safety and edge-case checklist

- Battery power whenever a person touches the electrodes; never mains. Keep the current near 0.1 mA at 5 kHz and confirm against IEC 60601-1 before any real use.
- **The arms and side paths send current across the chest.** Keep the ~0.1 mA limit in firmware (cap the DAC amplitude), and stop if the R5 reading shows more than 0.15 mA. 8-electrode body-composition scales work the same way, but our circuit is unverified.
- No pacemaker or defibrillator users. No magnets anywhere. All impedance off for implant patients.
- Round and insulate all metal edges. Keep the 18650 away from heat and metal; use a protected cell.
- Touch a grounded metal object before handling the ICs (static).
- Test with dry feet and hands only at first. Test wet feet after the circuit is verified.

Edge case	Prototype behaviour to demo
Guest on the stool	Weight outside the patient's band: shows "Is this Ramesh-ji?" and logs a guest
Wet feet, jewellery	Impedance shifts; the contact check prompts; the baseline learns the patient's usual jewellery
Loose grip on the handle	Z2 jumps; voice says "hold the handle firmly"; the trunk value becomes "no reading" for that day
Missing or low-quality reading	"No reading", never a guess
Missed day	Baseline kept; a reminder, then an SMS
Amulet forgotten	"No amulet data" flag; the stool still works
Power cut	Runs on the battery; messages queue
False alarm causes	Cause tags: infection-like, salty meal, missed medicine

10. Demo script (3 minutes), pitch and definition of done

1. **Story (20 s):** Ramaiah, 68, was admitted gasping at night. His weight had been rising for 9 days, and nobody knew.

2. **Sit on the stool (40 s):** weight, pulse, **leg fluid and chest fluid** appear, with a green light.
3. **Break it (80 s):** inject a guest, wet feet, a missed pill, an infection-like jump, and a real worsening (trunk impedance falling with weight rising). Show amber then red, the spoken reason, and the SMS on a phone.
4. **Results (30 s):** the four metrics against the weight-rule baseline, with limits stated.
5. **India and business (10 s):** offline, 2G SMS, ASHA loop, kit near cost plus a monthly care-loop service (hypothesis).

Pitch (5 min): story, problem, solution, live demo, results and honest limits, India fit and business, the ask. Record a backup demo video before the final run.

Done means: block diagram, schematic, LTspice plots, code on public GitHub with setup steps, results on 200+ virtual patients, 2-page report, pitch, disclaimer, and a working tier-A stool.

11. Not verified

- **All circuit values** (check in LTspice): INA126 gain and pin numbers, the 74HC4053 and **74HC4052** wiring, the ADC sampling rate, and the 5 kHz measurement accuracy.
- **Trunk impedance:** the subtraction formula, whether a 20–50 Ω trunk is resolvable, and whether it tracks lung fluid in real people.
- **Boot behaviour** with select lines on GPIO0 and GPIO12 on your exact board.
- **Prices:** about half the rows are estimates, and several ✓ items were out of stock when checked (INA126, 18650). Also unverified: the 2G modem on your SIM, and the pin map on your board.
- **Real-world behaviour:** amulet breathing in older people, pulse on cold or shaking fingers, and acceptance by elders.
- **Real people:** whether the impedance, ECG and four body paths behave as designed. Use phantoms and synthetic data for every result you report.